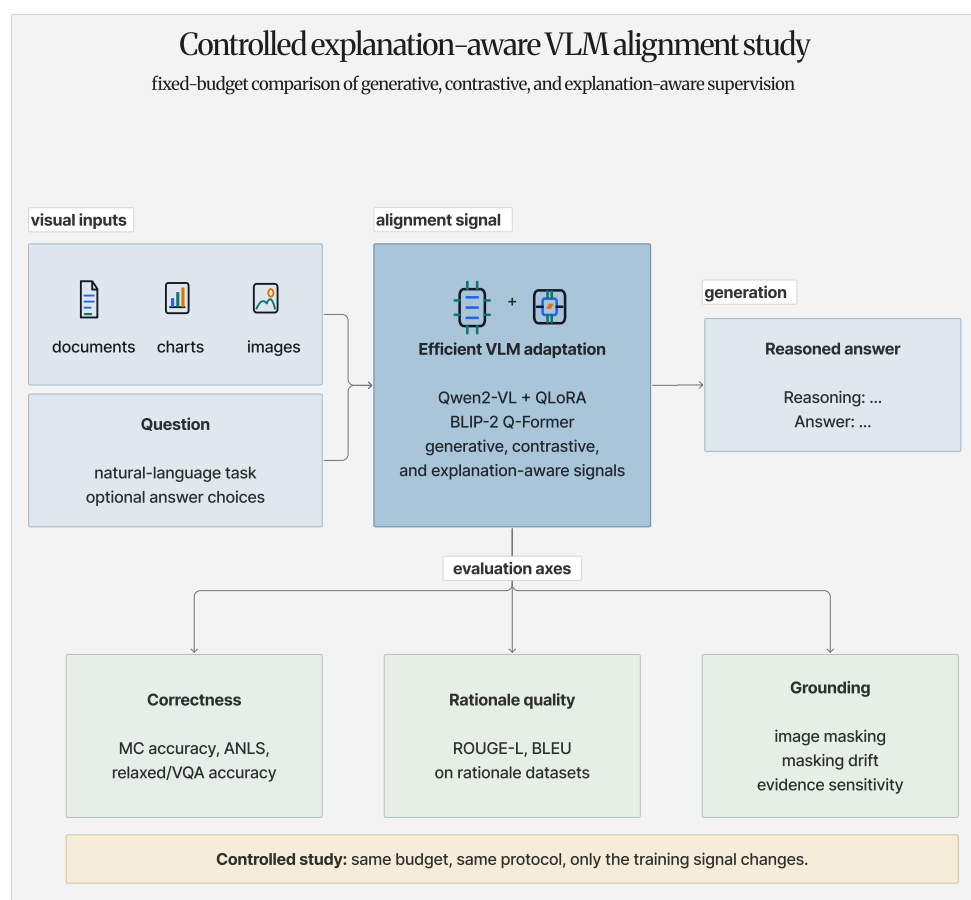


Graphical Abstract

Cross-Modal Generative Alignment for Document, Chart, and Image Reasoning with Explanation-Aware Vision–Language Models

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Highlights

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- The study runs zero-shot, answer-only, rationale-generative, explanation-aware, and BLIP-2 contrastive-enhanced controls under the same single-GPU environment.
- In rationale-bearing datasets, the explanation-aware training does help, but it’s not the best; answer-only controls are strong.
- BLIP-2 contrastive-enhanced fine-tuning yields only a small gain in ScienceQA with BLIP-2 over generative fine-tuning, and the retrieval diagnostic are very close to frozen and random controls.
- Evidence-masking drift shows evidence sensitivity of the model, so the answer gains, rationale quality, and grounding claims are kept separate.

Cross-Modal Generative Alignment for Document, Chart, and Image Reasoning with Explanation-Aware Vision–Language Models

Komron Valijonov^a, Raja Hashim Ali^{a,b}

^a*Department of Business, University of Europe for Applied Sciences, Think Campus, Konrad-Zuse-Ring 11, Potsdam, 14469, Brandenburg, Germany*

^b*Department of Artificial Intelligence, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, 23460, Pakistan*

Abstract

A vision-language model (VLM) is able to read a chart, document, scanned form, or natural image and answer questions about it. But how do we know if the answer is actually derived from the image itself, rather than being a lucky guess on a question based on the model’s remembered language patterns? The correctness of the answer itself is not enough if the reasoning for that answer is not grounded in the visual evidence of the image. Charts and documents are a good example of this problem, because to answer the question correctly, the model has to actually ground its reasoning on specific parts of the image, making it easier to spot when the model is looking at the wrong thing. Although recent work has made progress on this issues, there was no isolation of the training objective as the main variable, with all other factors such as the model, the data, and the budget held constant. Contrastive, generative, and explanation-based methods have all been proposed, but they have not been directly compared under the same conditions, so it is not clear what the objective itself contributed to the final performance. In this paper, we keep the pipeline fixed and change only the training signal, so we can see what the objective does to answer accuracy, explanation quality, and evidence sensitivity when the model, the data, and the compute budget are all held steady. We compare five settings head to head rather than treating them as unrelated systems: BLIP-2 generative fine-tuning, BLIP-2 contrastive-enhanced fine-tuning, Qwen2-VL under answer-only, rationale-generative, and explanation-aware supervision. All the runs use the same resource-constrained single-seed protocol, and the same eval-

uation code as much as possible, across ScienceQA, A-OKVQA, ChartQA, DocVQA, and a VQAv2 subset. The picture that emerges is mixed. On ScienceQA, Qwen2-VL-2B reaches 0.800 accuracy with answer-only supervision, 0.780 in both rationale-generative and length-aware explanation control settings, and 0.765 with fixed-alpha explanation-aware training. On A-OKVQA, the same model reaches 0.830 with answer-only supervision, 0.790 in both rationale-generative and length-aware explanation control training, and 0.795 with fixed-alpha explanation-aware training. ChartQA, DocVQA, and the VQAv2 subset have no gold rationales, so those runs fall back to answer-only controls; they lift the native headline metrics, but they do not test rationale supervision at all. BLIP-2 barely moves: the contrastive objective gives a small gain compared to the generative one on ScienceQA 0.480 vs 0.475, but the retrieval diagnostic stays near frozen and random controls in both cases. Scale is the exception. With a fixed explanation-aware objective on ScienceQA, Qwen2-VL-7B reaches 0.885 while the 2B version reaches 0.765. The masking analysis tells us whether the model relies on the image, not whether its explanation is faithful. Overall, the results do not show a clear win for explanation-aware training. The rationale supervision can help on a single-GPU budget, but the gain is easy to mis-credit to scale or tuning. Hence, it only stands up when answer-only controls, objective balance, model scale, transfer and evidence sensitivity are each reported separately.

Keywords: vision-language models, cross-modal alignment, explanation-aware training, chart and document understanding, faithfulness, parameter-efficient fine-tuning

1. Introduction

Vision-language models (VLMs) are becoming a default tool for answering questions based on the content of images, documents, and charts. The process looks simple: you give it an image and a question, and it will respond with an answer the image actually supports. What is considered as evidence depends entirely on the picture. In a document it might be sitting in a line of a body text, in a table, or in a stamped region of the scanned page. In a chart it might be in an axis label, the relative height of the bars, a colour, or a trend that is not written down explicitly in words. In science and commonsense visual question answering (VQA), it is often not in the image at all, forcing the model to deduce the answer from the background knowledge

it has already learned. Across every one of these, a correct-looking answer is not enough if the reasoning for that answer is not actually based on what is visible in the image. The VLMs are very good at generating the right-looking answers while looking at the wrong thing. That is why in this paper we do not rely on the answer alone, but also check the quality of the explanation and the sensitivity of the answer to visual evidence.

The issue behind all of this is a cross-modal alignment problem: how do we connect the visual signal to the language model in a way that actually makes the two talk to each other? It turns out that the quality of that wiring depends on many factors, such as the visual resolution, the data mixture, the connector design, the instruction tuning, and the adaptation method [1, 2, 3]. Open model families are very helpful, because backbones such as Qwen2-VL, BLIP-2, and LLaVA-OneVision are strong enough out of the box, while still being adaptable on an academic budget [4, 5, 6]. At the same time, document and chart studies show that general visual instruction following is not enough when it comes to text-heavy and structured images [7, 8, 9, 10].

Furthermore, there is also a budget reality to consider. Most students and small labs cannot afford full fine-tuning of a large multimodal model. Methods like DoRA (weight-decomposed low-rank adaptation) make it possible to fine-tune the model by updating only a small portion of the parameters, while keeping the base model frozen or quantised [11]. Hence, in this study, we take the same approach using low-rank adaptation (LoRA) and its quantised variant (QLoRA) instead of full-model fine-tuning [12, 13]. The main goal is not scoring the highest on a leaderboard, but rather to check the model’s performance when only the alignment objective changes while keeping all other factors constant.

1.1. Gap Analysis

The gap is isolation. Most papers change the model, the data, the training size, and the objective all at once. Once the final result is reported, it is hard to tell how much the objective itself contributed. Maybe the objective helped, but it might also be the case that it was simply a bigger model, cleaner data, or more tuning and not necessarily the objective itself.

Moreover, each objective has its own blind spot. Contrastive learning is good at matching an image to a caption, but it does not teach the model to generate an explanation or an answer from the image. Generative learning is good at producing an answer, but it does not necessarily mean that the

model is basing its answer on the correct parts of the image. Explanation-aware training teaches the model to actually write reasoning to come to the answer, but the model could have learned to write a convincing explanation after the fact, which might not actually be the cause of the answer.

Hallucination and faithfulness evaluation has mostly been studied on natural images, where the common test is whether the model names objects that are not there. Charts and documents have recently come into focus too, with benchmarks for chart hallucination, chart grounding, and document-answer groundedness emerging [14, 15, 16]. They are still comparatively underexplored, even though their evidence is more structured and easier to verify. That is exactly what makes them a good test case for grounding.

These gaps are not closed by a new model or benchmark. What is actually missing is a controlled study that holds the model, the data, the budget, and the evaluation procedure constant while changing only the alignment objective, to test not only the answer quality but also the explanation quality and the evidence sensitivity of the model under each objective.

1.2. Research Questions

The main question is to investigate how the resource-constrained vision-language model should be aligned to achieve both accurate and faithful visual reasoning. The following research questions break that down into more measurable parts.

1. RQ1: How have vision-language alignment methods evolved for grounded visual reasoning and explanation generation?
2. RQ2: How does BLIP-2 generative fine-tuning compare with BLIP-2 contrastive-enhanced fine-tuning on visual reasoning?
3. RQ3: Does explanation-aware fine-tuning improve answer accuracy and explanation quality on datasets with gold rationales?
4. RQ4: How does the same fine-tuning framework perform across document, chart, science, commonsense, and natural-image domains?
5. RQ5: How sensitive are the model’s answers to visual evidence under image-masking perturbations?
6. RQ6: How does the selected explanation-aware QLoRA setup scale from Qwen2-VL-2B to Qwen2-VL-7B?
7. RQ7: How well do fine-tuned adapters transfer across visual domains and prompt formats?

83 1.3. Problem Statement

84 The problem is cross-modal alignment in a resource-constrained environ-
85 ment. Given the same pretrained vision–language model, we analyse the
86 effect of different training objectives on answer accuracy, explanation qual-
87 ity, and evidence sensitivity. The model, the data format, the optimiser, and
88 the evaluation process are kept constant wherever possible, leaving only the
89 training signal as the variable. The BLIP-2 pair reuses one Q-Former train-
90 ing setup. The Qwen2-VL generative and explanation-aware runs reuse one
91 QLoRA setup. The goal is not to find the best scoring setup, but to observe
92 whether the chosen objective leads the model in a direction that matters for
93 a more trustworthy reasoning, and to check whether the answer quality, the
94 explanation quality, and the evidence sensitivity move together or separately
95 under each objective.

96 1.4. Novelty of this study

97 The contribution is the controlled comparison, not a new architecture.
98 Instead of introducing a new architecture, we compare existing ones under
99 different training objectives. We consider the faithfulness of the model’s
100 reasoning as a first-class result, rather than concentrating solely on answer
101 accuracy. The paper contributes the following:

- 102 • A comparison of zero-shot, answer-only, BLIP-2 generative, BLIP-2
103 contrastive-enhanced, Qwen2-VL rationale-generative, and Qwen2-VL
104 explanation-aware alignment under the same data and optimisation
105 settings.
- 106 • Evaluation along three separate axes, namely answer quality, explana-
107 tion quality, and evidence sensitivity, across document, chart, science,
108 and commonsense tasks.
- 109 • For multiple-choice reasoning, a generate-then-score protocol is used,
110 so the model is evaluated only after it has produced its own reasoning
111 trace.
- 112 • A direct test of how QLoRA behaves with scale, instead of assuming
113 that full fine-tuning or very large models are the only things worth
114 studying.

115 1.5. *Why this study matters*

116 The usefulness of this study is in the constraints. It runs on a resource-
117 constrained setup that is actually accessible to students and small labs. It
118 also avoids concentrating solely on a single aspect of performance, and in-
119 stead keeps answer accuracy, explanation quality, and evidence sensitivity as
120 separate readings. On documents and charts especially, that separation is
121 particularly important, because the evidence should trace back to a visible
122 region. The whole pipeline is reusable: every dataset is transformed into one
123 shared format, and every experiment is declared in a configuration file, so the
124 code can be used for future comparisons of new objectives, new models, or
125 new datasets, without worrying about hidden differences between runs. And
126 the results are honest rather than overselling: the rationale supervision does
127 help on the science and commonsense runs, but the answer-only controls stay
128 strong, and the BLIP-2 contrastive objective provides only a small answer
129 gain on top of weak retrieval evidence.

130 2. Literature Review

131 The review covers relatively latest works in the areas of vision–language
132 model design, text-rich and document-heavy images, charts and structured
133 visual reasoning, faithfulness and critique, and efficient adaptation. We group
134 it by the parts that actually matter here the most: model design, text-
135 rich images, chart reasoning, faithfulness, datasets, metrics, and low-budget
136 adaptation.

137 2.1. *Recent vision–language model design*

138 No recent VLM rides on a single alignment trick. What it does depends on
139 the backbone, the visual tokenisation, the data mixture, the connector, and
140 the instruction tuning, all interacting. Ablation work makes the case that
141 these knobs are still poorly separated, which is precisely the argument for
142 controlled comparisons [1]. MM1.5 drives it home: optical character recog-
143 nition (OCR) data, synthetic captions, instruction mixtures, and specialised
144 variants can each weigh as much as raw scale [2]. Molmo and PixMo are
145 useful for a different reason, namely open weights and open data that let you
146 actually read the recipe [3]. LLaVA-OneVision shows a single family stretch-
147 ing across image and video once the data mixture is chosen with care [6]. The
148 takeaway is blunt. Change the model, the data, and the budget together,
149 and the objective’s own contribution disappears into the noise.

150 Budget is the other reason open families matter. Qwen2-VL earns its
151 place by handling dynamic resolution, documents, charts, diagrams, and
152 other text-heavy inputs [4]. BLIP-2 earns its place differently: its Q-Former
153 is a convenient spot to bolt on a contrastive image-answer loss while the
154 heavy vision and language components stay frozen [5]. Neither is cited as
155 a leaderboard trophy. They are here as the kind of backbone that makes
156 parameter-efficient experiments possible at all.

157 *2.2. Text-rich and document-heavy images*

158 Documents are hard for a specific reason: the answer often hides in small
159 text, in the layout, or in local context. Reading and localising the evidence
160 is the task, not naming objects. A recent survey traces how quickly text-rich
161 image understanding has tilted toward multimodal large language models for
162 OCR-heavy and layout-heavy inputs [7]. DocLLM is a clean example, fold-
163 ing layout directly into the generative document-understanding problem [8].
164 OmniDocBench takes the coverage angle instead, scoring diverse PDF pars-
165 ing cases with detailed annotations [17]. Our document task is DocVQA [18].
166 We are after something narrower than document modelling at large: what
167 shifts when the objective changes and the rest of the training path holds
168 still?

169 *2.3. Charts and structured visual reasoning*

170 Charts break in their own way. Here the model has to read labels, make
171 sense of axes and tick marks, compare quantities, and now and then infer
172 a trend that no one spelled out in words. ChartInstruct and ChartGemma
173 show that instruction tuning on chart images lifts both comprehension and
174 reasoning [9, 10]. ChartMimic adds a useful twist through chart-to-code gen-
175 eration, which probes whether the model grasps the chart’s structure rather
176 than a single answer [19]. ChartMuseum is the sharper warning: on real-
177 world chart reasoning, current large vision-language models (LVLMs) still
178 sit well below humans [20]. MathVista, MMMU-Pro, and MMReason widen
179 the same gap into mathematical, expert-level, and multi-step multimodal rea-
180 soning [21, 22, 23]. That mix is why the design keeps ChartQA, ScienceQA,
181 and A-OKVQA: structured charts, science reasoning, and knowledge-heavy
182 visual reasoning in one sweep [24, 25, 26].

183 2.4. *Faithfulness, critique, and efficient adaptation*

184 Explanation-aware training carries one obvious risk: the model can learn
185 to write an explanation that reads well while ignoring the image entirely.
186 Recent benchmarks corner this from several sides. MMMU probes expert-
187 level perception and reasoning across many visual formats [27]. MMStar
188 asks the awkward question of whether an example is truly vision-dependent
189 or solvable from language priors alone [28]. HallusionBench zeroes in on
190 where language hallucination meets visual illusion in LVLM answers [29].
191 Hallucination-augmented contrastive learning reframes hallucinated visual
192 text as a representation-learning problem [30]. VISCO and Critic-V push
193 past final-answer accuracy into critique, correction, and error detection [31,
194 32]. Read together, they make one case: measure faithfulness apart from
195 accuracy.

196 Efficient adaptation is the hard constraint we live under. DoRA is the
197 recent example we point to, sharing the same motive: adapt well without
198 paying for full-model fine-tuning [11]. Reason-RFT gestures the other di-
199 rection we care about, toward post-training that explicitly rewards visual
200 reasoning [33]. Our own work stays with supervised objective comparisons,
201 but the same worry sits behind it. We judge the model as a reasoning system,
202 not merely an answer generator.

203 Table 1 lays the closest lines of work beside the gap we target. It is not
204 meant as a complete catalogue. The point is narrower: a fixed-budget objec-
205 tive comparison still earns its place even when stronger specialised models
206 already exist.

207 One pattern runs through all of it. Early multimodal alignment mostly
208 meant learning a shared image-text representation, or generating a short
209 answer and stopping there. The recent wave pushes much further: harder
210 visual reasoning, instruction tuning, text-rich images, critique, hallucination
211 checks, efficient adaptation. What still goes unmeasured is the objective on
212 its own, namely how much it moves answer quality, rationale quality, and
213 visual evidence use once the model, the data cap, and the compute budget
214 are nailed down. That measurement is the job of the experiments below.

215 3. Methodology

216 One rule drives the method: hold the pipeline fixed, change the objective.
217 A dataset goes in and is converted to a shared format. From there the model
218 is loaded, QLoRA adapters are attached if the run needs them, the chosen

Table 1: Summary of representative related work and the gap addressed by this study. A check mark indicates that the paper directly targets the property, a cross indicates that it is outside the main scope, and a tilde indicates partial coverage.

Work	Main focus	Generation	Structured VQA	Rationales	Efficient FT	Relation to this study
Recent VLM construction [1, 3, 6, 2]	Backbone and data recipe	✓	~	~	~	Shows why architecture and data must be controlled before claiming an objective-level effect.
Open VLM backbones [4, 5]	Adaptable model families	✓	~	~	~	Provide practical backbone candidates for fixed-budget experiments.
Text-rich document work [7, 8, 17]	Documents and PDF parsing	✓	✓	~	~	Motivates document evaluation, but does not isolate training objectives under one backbone.
Chart reasoning work [9, 10, 20, 19]	Chart understanding	✓	✓	~	~	Shows that chart reasoning remains difficult and needs explicit structured evaluation.
Recent reasoning benchmarks [21, 27, 28, 22, 23]	Harder multi-modal evaluation	~	✓	~	✗	Motivate stricter evaluation for visual, mathematical, and multi-step reasoning.
Faithfulness and critique [29, 30, 31, 32]	Reasoning reliability	~	✓	✓	✗	Supports measuring hallucination, critique, and evidence use separately from accuracy.
DoRA / Reason-RFT [11, 33]	Efficient and reasoning-focused adaptation	~	✗	~	✓	Motivate low-resource adaptation and explicit reasoning-oriented post-training.
This study	Controlled explanation-aware alignment	✓	✓	✓	✓	Isolates objective choice and measures answer, rationale, and evidence-sensitivity outcomes.

objective is trained, and evaluation goes through the same generate-then-score procedure every time. Figure 1 lays out the full pipeline. The design is conservative on purpose. For what we are asking, one clean controlled comparison beats a sprawling leaderboard-style run. Every experiment is declared in a configuration file: the dataset, the backbone, the objective, the adapter settings, the metrics, and the seed, all on the record. One implementation path handles baseline evaluation, fine-tuning, and final evaluation. That is what keeps hidden differences from creeping in between runs.

3.1. Dataset

Five public benchmarks carry the experiments. DocVQA stands in for documents, where the answer usually rides on scanned text, layout, and local visual context [18]. ChartQA covers chart reasoning, where it can turn on a label, an axis, a value, or a comparison [24]. ScienceQA brings multimodal science questions with answer choices and written explanations, which is what makes it usable for rationale supervision [25]. A-OKVQA adds commonsense questions and rationales over natural images [26]. VQAv2 plays the general natural-image control [34]. One caveat there: the completed run draws on the configured `Erland/VQAv2-sample` source, capped at 900 training and 100 evaluation examples, so we report it as a subset result, not a full VQAv2 score. Whatever the source, every dataset is bent into one schema: image, question, optional answer choices, target answer, optional

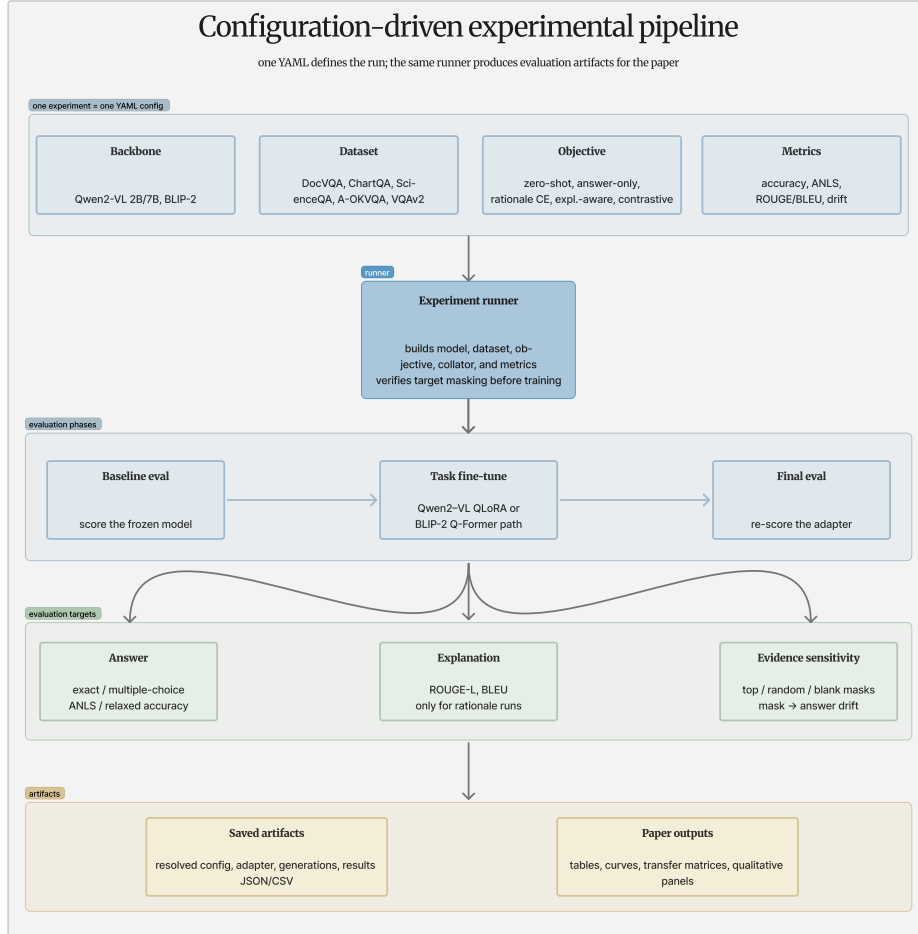


Figure 1: Overall workflow of the proposed explanation-aware alignment study. Datasets are normalised into a shared multimodal example format. Qwen2-VL runs are adapted with QLoRA, BLIP-2 runs train the Q-Former-side alignment path, and each objective is evaluated through answer, rationale, and evidence-sensitivity metrics.

240 rationale. A dataset with no gold rationale is treated as answer-only, unless
 241 a separate rationale experiment is explicitly marked.

Table 2: Benchmarks used in the study. The sizes are rounded public benchmark sizes before filtering by image availability, answer type, or rationale availability. The completed experiments use capped train and evaluation splits so that every run fits the same practical compute budget.

Dataset	Domain	Approx. size	Answer type	Gold rationale	Primary metric
DocVQA [18]	Document	50K	short text span	no	ANLS
ChartQA [24]	Chart	32K	numeric / short text	no	relaxed accuracy
ScienceQA [25]	Science image	21K	multiple choice	yes	accuracy
A-OKVQA [26]	Commonsense image	25K	multiple choice / short text	yes	accuracy
VQAv2 subset [34]	Natural image	900/100 run split	open ended	no	VQA accuracy

Table 3: Input construction examples. The complete token, label, span, and collator traces are saved as pipeline artifacts; this table gives the paper-level view of what each prompt family means.

Prompt family	Raw fields	Prompt seen by model	Supervised target	Supervised spans
Answer-only	image, question, answer	Question: ... plus choices when available	Answer: <answer>.	answer tokens only
Rationale-generative	image, question, answer, gold rationale	Question: ... plus choices when available	Reasoning: <rationale> Answer: <answer>.	full target under uniform CE
Explanation-aware	image, question, answer, gold rationale	Question: ... plus choices when available	Reasoning: <rationale> Answer: <answer>.	rationale and answer spans weighted separately
Answer-only fall-back	image, question, answer, no gold rationale	Question: ...	Answer: <answer>.	answer tokens only

3.2. Detailed methodology

Figure 2 shows the model-level setup. Every backbone sits behind one common interface, so the training code never forks into a separate path per model. Qwen2-VL loads in 4-bit precision and picks up rank-16 QLoRA adapters [12, 13]. On the BLIP-2 contrastive arm, the vision encoder and the OPT (Open Pre-trained Transformer) language model are frozen solid. Only the Q-Former-side components, the language projection, and a small answer-projection head ever update. The baseline variant just evaluates the model before any task fine-tuning. The contrastive-enhanced variant bolts on an auxiliary image-answer term: it tugs the pooled Q-Former image representation toward the correct answer and away from the others in the batch. The generative variant trains the model to produce the answer from the image and question. The explanation-aware variant trains it to produce a rationale first, then the answer. So the variants differ in one thing only, the supervision signal, and everything else inside a comparison is held as close as it can be.

In generative training, prompt and target live in one concatenated sequence, but the loss only ever touches the target tokens. That detail is easy to get wrong, because multimodal processors slip image placeholder tokens in front of the text. Count the prompt length by hand and the supervised span

262 quietly slides off position. The implementation sidesteps it by measuring
 263 prompt length with the very same processor that encodes the full sequence.
 264 The contrastive objective reuses those span tags, pooling answer tokens alone
 265 into the InfoNCE (information noise-contrastive estimation) text representa-
 266 tion [35]. Rationale tokens still receive the generative loss; they simply are
 267 not treated as the contrastive positive. For the explanation-aware objective,
 268 the target splits in two, an explanation span and an answer span. Equation 1
 269 weights the two spans with α . When $\alpha = 1$, only the answer span is weighted,
 270 although the target sequence can still contain the gold rationale before the
 271 answer. A clean answer-only control is therefore reported separately, using
 272 a target template that contains only **Answer:**. When $\alpha = 0$, only the ex-
 273 planation span is supervised. Intermediate values test whether answer and
 274 rationale supervision help each other or compete.

$$\mathcal{L} = \alpha \mathcal{L}_{\text{answer}} + (1 - \alpha) \mathcal{L}_{\text{explanation}} \quad (1)$$

275 where $\mathcal{L}_{\text{answer}}$ and $\mathcal{L}_{\text{explanation}}$ are the masked cross-entropy losses over the
 276 answer and explanation token spans, and α controls their relative weight.
 277 There are two alpha modes. Fixed-alpha mode treats α as a dial we set
 278 by configuration, an ablation knob. Length-aware mode does something
 279 different: it derives the effective answer weight straight from the supervised
 280 token counts:

$$\alpha_{\text{eff}} = \frac{\lambda n_{\text{answer}}}{\lambda n_{\text{answer}} + n_{\text{explanation}}}, \quad (2)$$

281 where n_{answer} and $n_{\text{explanation}}$ are the answer and explanation token counts
 282 after collation, and λ is the answer-weight multiplier. With $\lambda = 1$, this
 283 collapses to plain uniform token-level cross-entropy over the rationale-plus-
 284 answer target. So the fixed-alpha sweep is the ablation; the length-aware run
 285 is the audit control for the natural, token-weighted setting.

286 Multiple-choice datasets are scored in two steps at inference. First the
 287 model writes a short reasoning trace from the image and question. Then,
 288 in the context of that trace, each candidate answer gets a length-normalised
 289 conditional likelihood, and the highest score wins. The two-step design is
 290 there to avoid brittle string parsing and option-label copying. Open-ended
 291 datasets are simpler: the generated answer string is scored with the dataset’s
 292 native metric. Whenever the model produces a rationale, we save it, and
 293 those saved rationales feed the text-quality and faithfulness analysis.

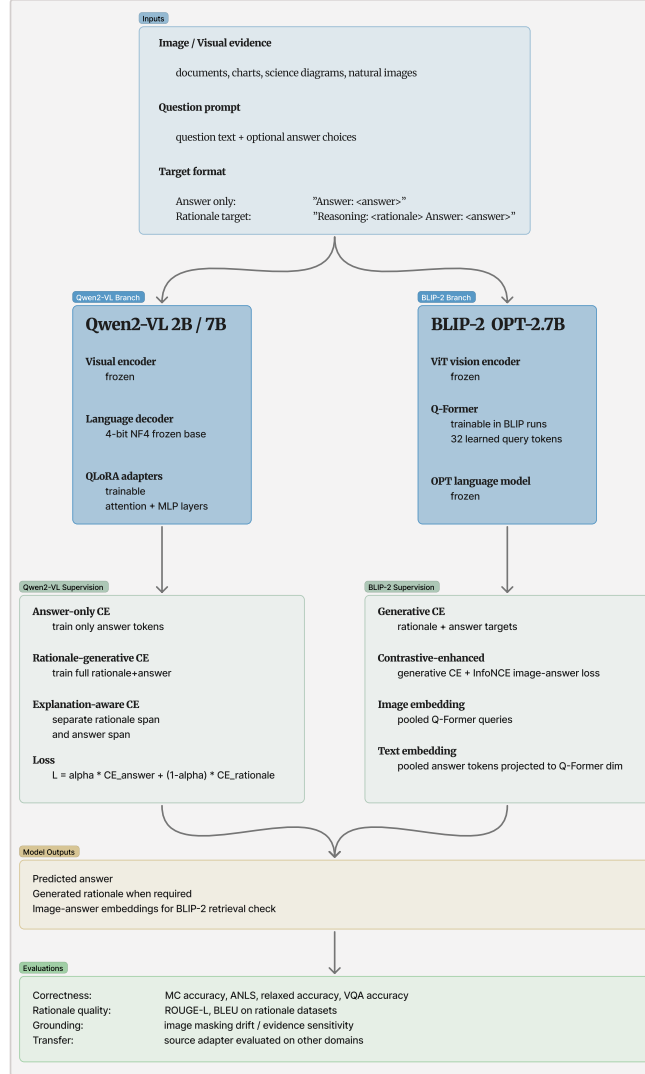


Figure 2: Model-level architecture for explanation-aware cross-modal alignment. Visual evidence and the question prompt are encoded into a multimodal token stream. The Qwen2-VL path uses trainable QLoRA adapters and separate rationale and answer supervision spans. The BLIP-2 path keeps the vision tower and OPT language model frozen and trains the Q-Former-side generative or contrastive alignment components.

294 3.3. Evaluation metrics

295 Evaluation splits into three parts: answer quality, rationale quality, and
 296 evidence sensitivity. Answer quality always uses the benchmark’s own na-

297 tive metric. When the tables say “headline metric,” they mean exactly that
 298 native primary metric for the target dataset, not some universal number to
 299 be read straight across datasets. For ScienceQA and A-OKVQA that metric
 300 is multiple-choice accuracy. DocVQA uses average normalised Levenshtein
 301 similarity (ANLS), per the benchmark convention [18]. ChartQA uses re-
 302 laxed accuracy on numerical answers [24]. The VQAv2 subset uses VQA
 303 consensus accuracy wherever annotator answers exist [36, 34]. Rationale
 304 quality leans on BLEU and ROUGE-L whenever gold rationales are avail-
 305 able [37, 38]. Neither is a perfect judge, but together they give a basic lexical
 306 and sequence-level check on the explanation. Evidence sensitivity asks a
 307 blunter question: does the answer’s likelihood move when we mask part of
 308 the image? This borrows the old idea of occlusion sensitivity analysis [39].
 309 The specific measure is evidence-masking consistency, written out in Equa-
 310 tion 3. The qualitative diagnostic then lines up the high-drift regions against
 311 whatever visual evidence the generated explanation actually names, when
 312 that comparison is possible.

$$\text{drift}(r) = s(a \mid I) - s(a \mid I \setminus r) \quad (3)$$

313 where $s(a \mid I)$ is the length-normalised log-likelihood of the gold answer
 314 a given image I , and $I \setminus r$ is the same image with region r occluded. A
 315 large positive drift says the masked region mattered. A drift near zero says
 316 the opposite: the answer probably leaned on language priors, a shortcut, or
 317 non-visual context. The absolute scale does not travel across datasets, since
 318 answer length, prompt family, image resolution, and task format all bend the
 319 likelihood scores. The split it captures is the familiar one between a useful
 320 explanation and a faithful one. It also speaks to recent hallucination and
 321 critique work, where a model happily describes or relies on visual content
 322 that simply is not there [29, 30, 31, 32]. That is why evidence sensitivity
 323 sits next to answer accuracy here, rather than being parked as a post-hoc
 324 example.

325 3.4. *Experimental settings*

326 Every variant runs under the same fixed-budget protocol, and any de-
 327 viation from it is written into the experiment ledger. The Qwen2-VL runs
 328 share one recipe: AdamW, a learning rate of 2×10^{-4} , cosine decay, three
 329 percent warm-up, weight decay of 0.01, bf16 compute, and an effective batch
 330 size of 32 reached through gradient accumulation. The base model loads

331 under 4-bit NormalFloat (NF4) quantisation. Unless a result table says oth-
 332 erwise, the QLoRA adapters use rank 16 and scaling factor 32 [12, 13]. The
 333 workhorse backbone is Qwen2-VL-2B-Instruct, strong enough for the target
 334 tasks yet small enough to fit the budget [4]. For BLIP-2 we use Salesforce
 335 BLIP-2 OPT-2.7B, vision tower and OPT language model frozen, training
 336 only the Q-Former-side alignment components [5]. The scale ablation reuses
 337 the same QLoRA recipe on Qwen2-VL-7B. Each run logs its dataset split, fil-
 338 tered count, seed, objective, adapter configuration, and metrics. The pipeline
 339 also writes a split-and-leakage audit from the selected train and evaluation
 340 identifiers, wherever the dataset exposes stable IDs. For the capped datasets
 341 in the final artifact set, that overlap count comes back at zero. DocVQA is
 342 the one non-standard split: the loaded dataset only exposes validation and
 343 test, so the controlled run trains on `validation[:80%]` and evaluates on
 344 `validation[80%:]`, with native-ID overlap checking. The audit shows zero
 345 native-ID overlap, but it cannot prove document-level disjointness past the
 346 IDs the loader exposes. So we treat the DocVQA number as a capped in-
 347 ternal adaptation result, not a public-test benchmark claim. We also attach
 348 approximate uncertainty estimates to the accuracy-style headline metrics.
 349 With most evaluation caps at 200 examples and only a few seeds in the bud-
 350 get, a gap like 0.790 versus 0.795 is an observed difference, nothing more.
 351 Table 4 lists the default configuration.

Table 4: Default Qwen2-VL QLoRA fine-tuning configuration. BLIP-2 contrastive runs
 use the separately reported frozen-vision/frozen-LLM Q-Former training setup. Any run
 that changes these values is reported separately in the experiment ledger and in the Results
 section.

Training Configuration	
Adapter	QLoRA (4-bit NF4, rank 16, scaling 32)
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)
Learning rate	2×10^{-4} (cosine, 3% warm-up)
Weight decay	0.01
Effective batch size	32 with gradient accumulation
Precision	bf16 compute
Primary seed policy	one seed per complete configuration

352 *3.5. Experiment ledger*

353 Before any scores, Table 5 lays out the comparisons. The line that mat-
354 ters is rationale availability. ScienceQA and A-OKVQA can carry rationale-
355 generative and explanation-aware supervision, because they ship gold ratio-
356 nales. ChartQA, DocVQA, and VQAv2 stay on answer-only fallback, plus
357 the transfer and domain runs, until some future experiment adds rationale
358 supervision of their own.

Table 5: Main experiment ledger. The full generated ledger includes exact caps, seeds, prompt variants, alpha mode, and configuration hashes.

Comparison	Runs included	Use in the paper
BLIP-2 objective pair	BLIP-2 OPT-2.7B on ScienceQA: generative and generative+contrastive.	Tests whether the auxiliary contrastive objective adds anything beyond BLIP-2 generative fine-tuning.
ScienceQA controls	Qwen2-VL-2B on ScienceQA: answer-only, rationale+answer CE, fixed- α explanation-aware sweep, and length-aware control.	Separates answer supervision, rationale supervision, and alpha weighting on a gold-rationale dataset.
A-OKVQA controls	Qwen2-VL-2B on A-OKVQA: answer-only, rationale+answer CE, fixed- α explanation-aware, and length-aware control.	Repeats the rationale-supervision comparison on a second gold-rationale domain.
Fallback domains	Qwen2-VL-2B on ChartQA, DocVQA, and VQAv2 with answer-only fallback prompts.	Reports in-domain adaptation and transfer for datasets without gold rationales; these rows are not used as explanation-aware evidence.
Scale	Qwen2-VL-2B and Qwen2-VL-7B on ScienceQA with the same fixed explanation-aware QLoRA setting.	Checks whether the selected setup changes when the backbone scale increases.
Transfer	Qwen2-VL-2B source adapters evaluated on target-domain prompts across all datasets.	Tests cross-domain behaviour separately from the official in-domain result table.
Masking	Selected Qwen2-VL runs with image-region perturbation and saved visual examples.	Measures evidence sensitivity beside answer accuracy and rationale quality.

359 4. Results

360 The results track the comparison types from the ledger, in order. The
361 BLIP-2 objective pair comes first, testing whether the auxiliary contrastive
362 term earns its place beyond generative fine-tuning. Then the ScienceQA
363 and A-OKVQA runs pull answer-only, rationale-generative, and explanation-
364 aware supervision apart on the two datasets that have gold rationales. The
365 chart, document, and VQAv2 runs follow, showing in-domain adaptation
366 where no gold rationales exist. After that come masking-based evidence sen-
367 sitivity, the Qwen2-VL scale check, and cross-domain transfer. One reminder
368 throughout: every score comes from the capped single-seed protocol above,
369 so they are controlled framework results, not leaderboard numbers.

370 4.1. Objective-level comparison

371 Table 6 holds the direct objective comparisons. Take the BLIP-2 Sci-
372 enceQA pair first: both trained variants clear the 0.445 frozen baseline, the
373 contrastive-enhanced run landing at 0.480 and the generative control at 0.475.
374 That gap is tiny, and it sits inside the approximate uncertainty intervals in
375 Table 7. Retrieval tells the same cautious story: the contrastive-enhanced
376 model posts recall@1 (R@1) of 0.010, R@5 of 0.020, R@10 of 0.030, and a
377 mean reciprocal rank (MRR) of 0.0307. Qwen2-VL-2B is stronger across
378 the board. Rationale-generative fine-tuning reaches 0.780 on ScienceQA and
379 0.790 on A-OKVQA. On A-OKVQA the explanation-aware run reaches 0.795
380 accuracy, 0.337 ROUGE-L, and 0.087 BLEU, numerically a hair above the
381 generative control, which we read as an observed single-run difference rather
382 than a settled win. The honest surprise is the answer-only controls: 0.800
383 on ScienceQA and 0.830 on A-OKVQA, both strong. The zero-shot col-
384 umn, meanwhile, is one shared backbone-dataset reference for orientation,
385 not a fresh prompt-family baseline per row. Back on ScienceQA, the fixed
386 $\alpha = 0.50$ explanation-aware run reaches 0.765. The length-aware control
387 reaches 0.780, level with the rationale-generative control but still under the
388 answer-only number.

Table 6: Objective-level results under the controlled protocol. The zero-shot reference is the shared pre-fine-tuning score for the same backbone and dataset group, not a separately recomputed frozen score for every prompt family. Accuracy is multiple-choice accuracy for ScienceQA and A-OKVQA.

Backbone	Dataset	Objective	Zero-shot ref.	Final	ROUGE-L	BLEU
BLIP-2 OPT-2.7B	ScienceQA	Generative	0.445	0.475	–	–
BLIP-2 OPT-2.7B	ScienceQA	Contrastive-enhanced	0.445	0.480	–	–
Qwen2-VL-2B	ScienceQA	Answer-only	0.400	0.800	–	–
Qwen2-VL-2B	ScienceQA	Rationale+answer CE	0.400	0.780	0.693	0.559
Qwen2-VL-2B	ScienceQA	Explanation-aware, $\alpha = 0.50$	0.400	0.765	0.643	0.493
Qwen2-VL-2B	ScienceQA	Explanation-aware, length-aware	0.400	0.780	0.696	0.563
Qwen2-VL-2B	A-OKVQA	Answer-only	0.355	0.830	–	–
Qwen2-VL-2B	A-OKVQA	Rationale+answer CE	0.355	0.790	0.329	0.083
Qwen2-VL-2B	A-OKVQA	Explanation-aware, $\alpha = 0.50$	0.355	0.795	0.337	0.087
Qwen2-VL-2B	A-OKVQA	Explanation-aware, length-aware	0.355	0.790	0.330	0.083

Table 7: Approximate uncertainty for selected accuracy-style headline metrics. These intervals use the generated normal-approximation standard errors for capped evaluation sets, so they are a caution about small observed differences rather than a replacement for multi-seed testing.

Run	Metric	Eval cap	Score	Approx. 95% CI
BLIP-2 generative ScienceQA	MC accuracy	200	0.475	0.406–0.544
BLIP-2 contrastive ScienceQA	MC accuracy	200	0.480	0.411–0.549
ScienceQA answer-only	MC accuracy	200	0.800	0.745–0.855
ScienceQA rationale+answer CE	MC accuracy	200	0.780	0.723–0.837
ScienceQA explanation-aware, $\alpha = 0.50$	MC accuracy	200	0.765	0.706–0.824
ScienceQA explanation-aware, length-aware	MC accuracy	200	0.780	0.723–0.837
A-OKVQA answer-only	MC accuracy	200	0.830	0.778–0.882
A-OKVQA rationale+answer CE	MC accuracy	200	0.790	0.734–0.847
A-OKVQA explanation-aware, $\alpha = 0.50$	MC accuracy	200	0.795	0.739–0.851
Qwen2-VL-7B ScienceQA, $\alpha = 0.50$	MC accuracy	200	0.885	0.841–0.929

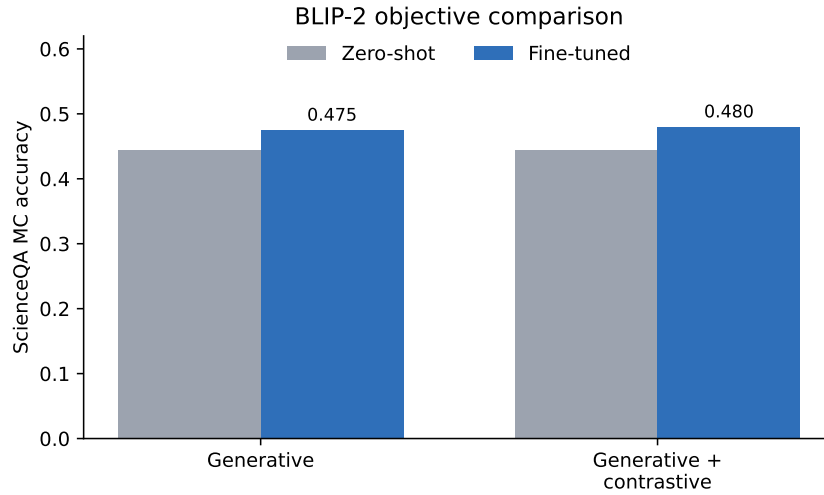


Figure 3: BLIP-2 generative and contrastive-enhanced ScienceQA results. Both runs improve over the frozen baseline, with only a small observed gain from the auxiliary contrastive term.

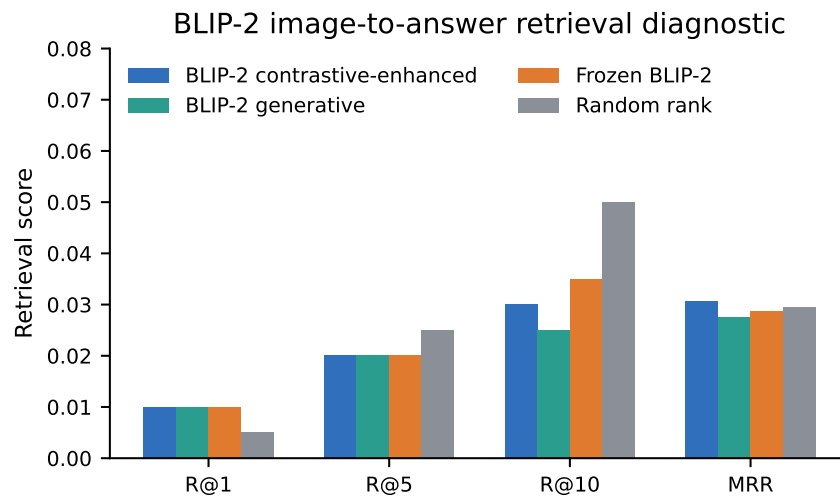


Figure 4: BLIP-2 image-to-answer retrieval diagnostic. The analysis reports frozen BLIP-2, BLIP-2 generative, BLIP-2 contrastive-enhanced, and random-rank controls so the contrastive branch is not interpreted without a reference point.

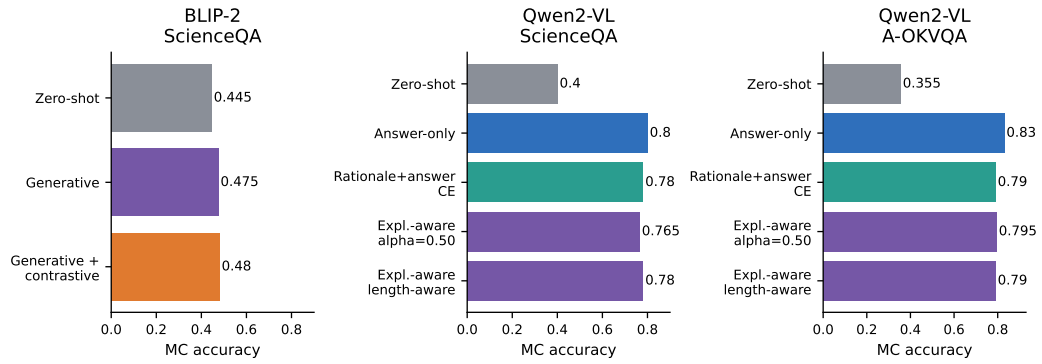


Figure 5: Controlled strategy comparison for BLIP-2 and Qwen2-VL. The figure shows zero-shot, answer-only, rationale-generative, explanation-aware, and contrastive variants where those controls are available.

389 *4.2. Effect of explanation-aware training*

390 Table 8 and Figure 6 trace the ScienceQA α -sweep, where α is the answer-
 391 span weight from Equation 1. Fine-tuning alone lifts accuracy from the
 392 0.400 zero-shot reference into the 0.690–0.780 band. The sweet spot is low:
 393 $\alpha = 0.10$, at 0.780. Push α toward the answer-heavy end and accuracy slides,
 394 to 0.690 at $\alpha = 0.75$ and 0.695 at $\alpha = 1.00$. The explanation metrics move
 395 in lockstep. ROUGE-L and BLEU also peak at $\alpha = 0.10$ (0.695 and 0.566)
 396 and decay as α climbs. At $\alpha = 1.00$ only the answer span is supervised
 397 inside a rationale-shaped target, so there are no rationale metrics to report.
 398 In accuracy, the $\alpha = 0.10$ run ties the rationale-generative and length-aware
 399 controls. One trap worth flagging: the $\alpha = 1.00$ row is not the true answer-
 400 only control, because the model still sees a rationale-shaped template even
 401 though the rationale span goes unsupervised. The genuine answer-only Sci-
 402 enceQA control, on its own prompt family, reaches 0.800. And all of this
 403 rests on one seed, 2,000 training and 200 evaluation examples, so these are
 404 observed effects, not final statistical proof.

Table 8: Explanation-aware fixed- α sweep on ScienceQA (Qwen2-VL-2B, QLoRA, 2,000 train / 200 eval, single seed). The coefficient α weights the answer span in Equation 1: $\alpha = 1$ is answer-span-only within the rationale target and $\alpha = 0$ is explanation-span-only. The zero-shot and rationale-generative rows are references. Random-choice accuracy is 0.354. The best fixed α is shown in bold.

Variant (α)	MC accuracy	ROUGE-L	BLEU
Zero-shot (frozen)	0.400	–	–
Rationale+answer CE	0.780	0.693	0.559
$\alpha = 0.00$ (explanation-only)	0.725	0.417	0.299
$\alpha = 0.10$	0.780	0.695	0.566
$\alpha = 0.25$	0.775	0.686	0.548
$\alpha = 0.50$	0.765	0.643	0.493
$\alpha = 0.75$	0.690	0.604	0.443
$\alpha = 1.00$ (answer-span only)	0.695	–	–

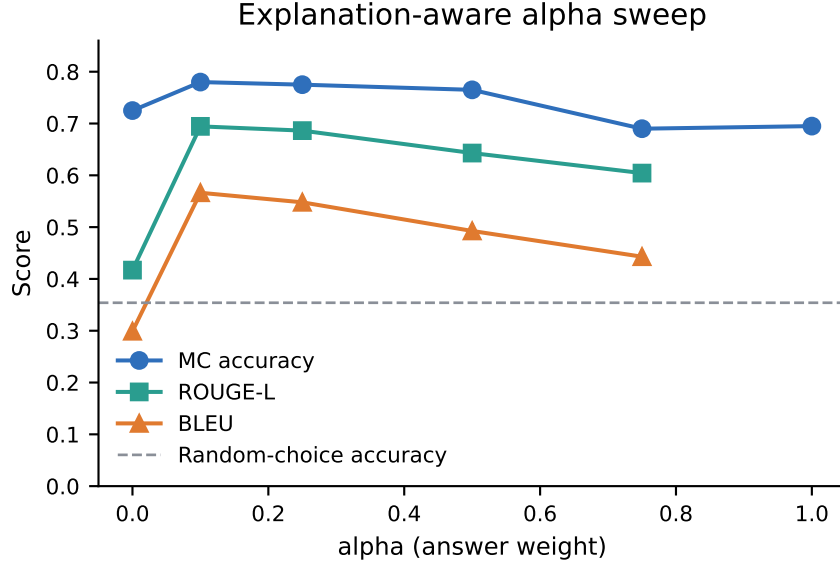


Figure 6: Fixed-alpha explanation-aware sweep on ScienceQA. Answer accuracy peaks at the low-answer-weight setting and declines toward both extremes inside the rationale-plus-answer target. Explanation quality (ROUGE-L, BLEU) peaks at the same point and falls as α increases. Series are distinguished by marker shape and line style; the dashed grey line marks random-choice accuracy. Single seed.

4.3. Domain-level behaviour

Table 9 and Figure 7 gather the selected domain runs from the final matrix. For ScienceQA the row uses the fixed $\alpha = 0.50$ explanation-aware run, so it lines up with the in-domain figure; the stronger $\alpha = 0.10$ and length-aware settings live in the sweep table above. The biggest jumps land on DocVQA, VQAv2, ChartQA, and A-OKVQA. A-OKVQA climbs from 0.355 to 0.795 accuracy against the reported zero-shot reference. The VQAv2 subset goes from 0.350 to 0.730 VQA accuracy. ChartQA reaches 0.655 relaxed accuracy, exact match 0.410. DocVQA is the headline, at 0.905 ANLS with exact match at 0.855. That number is strong for a capped internal run, but it deserves caution on three counts: the qualitative samples are short extractive answers, the setup is answer-only fallback rather than gold rationales, and the split audit cannot prove document-level disjointness past the native IDs it can see.

Table 9: Domain-level results for the completed Qwen2-VL-2B runs. Each domain uses its native headline metric: multiple-choice accuracy for ScienceQA and A-OKVQA, relaxed accuracy for ChartQA, ANLS for DocVQA, and VQA accuracy for the VQAv2 subset.

Domain	Dataset	Objective	Zero-shot ref.	Final	Secondary metric
Science	ScienceQA	Explanation-aware, $\alpha = 0.50$	0.400	0.765	ROUGE-L 0.643, BLEU 0.493
Commonsense	A-OKVQA	Explanation-aware, $\alpha = 0.50$	0.355	0.795	ROUGE-L 0.337, BLEU 0.087
Charts	ChartQA	Answer-only fallback	0.335	0.655	Exact match 0.410
Documents	DocVQA	Answer-only fallback	0.349	0.905	Exact match 0.855
Natural images	VQAv2 subset	Answer-only fallback	0.350	0.730	Exact match 0.350 $\rightarrow$ 0.730

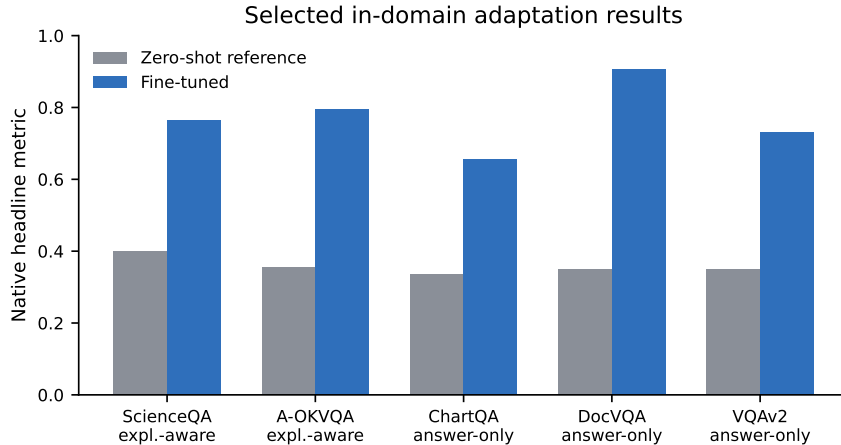


Figure 7: In-domain adaptation results for the selected completed Qwen2-VL-2B domain runs. The comparison mixes rationale-supervised ScienceQA/A-OKVQA rows with answer-only fallback rows for domains without gold rationales, so it should not be read as an explanation-aware effect across all domains.

4.4. Faithfulness and hallucination analysis

Table 10 and Figure 8 carry the evidence-masking drift check. Each number is a mean over 30 evaluation examples. The bigger and more positive it is, the more masking a region costs the gold-answer likelihood, which means the model was relying on what we hid. The drift piles up on the structured tasks: DocVQA at 0.375, ChartQA at 0.224, VQAv2 at 0.111. ScienceQA and A-OKVQA usually sit far lower, with one loud exception, the ScienceQA answer-only adapter at 0.0898. That low ScienceQA drift is not surprising; the questions come loaded with language priors and explicit answer choices, the visual evidence is diffuse, and a coarse mask can miss the single cue the model actually used. Explanation supervision does not move drift in any clean way here. At $\alpha = 0.10$ it is 0.0223, at $\alpha = 0.50$ it is 0.0197, and answer-span-only $\alpha = 1.00$ gives 0.0149, despite that run producing no supervised

432 rationale at all. On A-OKVQA the explanation-aware run drifts a touch
433 more than the generative control, 0.0271 against 0.0245, but the intervals
434 around both are wide. Read plainly, the masking result backs a domain-
435 sensitivity story, not the sweeping claim that explanation-aware training by
436 itself buys faithfulness. We report no manual hallucination rate, because we
437 never ran a fixed annotation protocol for hallucination categories.

Table 10: Evidence-masking drift by completed run. Higher values indicate stronger sensitivity to masked visual evidence. The intervals are approximate 95% intervals from the generated masking artifact.

Run	Mean drift	Approx. 95% CI	Examples
ScienceQA answer-only	0.0898	0.0197–0.1830	30
ScienceQA rationale+answer CE	0.0217	-0.0100–0.0582	30
ScienceQA $\alpha = 0.10$	0.0223	-0.0080–0.0568	30
ScienceQA $\alpha = 0.50$	0.0197	-0.0119–0.0552	30
ScienceQA answer-span-only $\alpha = 1.00$	0.0149	-0.0274–0.0638	30
A-OKVQA rationale+answer CE	0.0245	-0.0236–0.0726	30
A-OKVQA explanation-aware	0.0271	-0.0075–0.0627	30
ChartQA answer-only fallback	0.2237	0.1419–0.3140	30
DocVQA answer-only fallback	0.3749	0.2770–0.4960	30
VQAv2 answer-only fallback	0.1109	0.0274–0.2408	30
Qwen2-VL-7B ScienceQA $\alpha = 0.50$	0.0114	-0.0059–0.0260	30

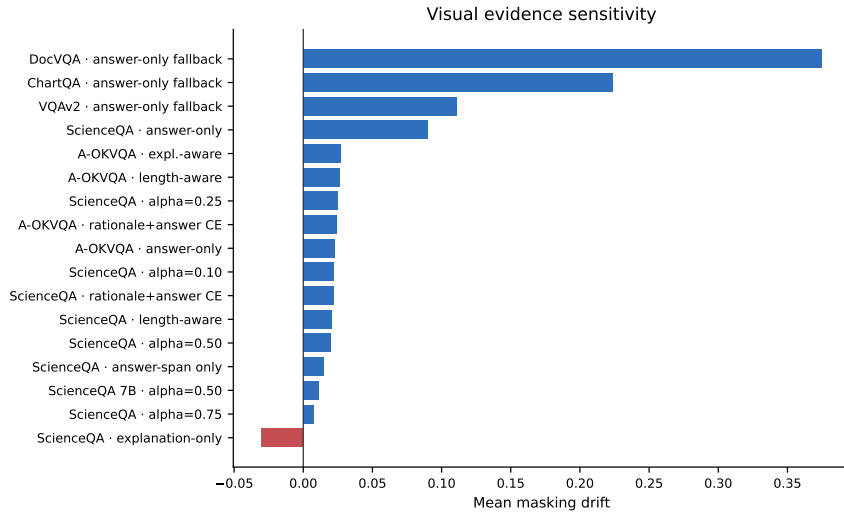


Figure 8: Mean evidence-masking drift across completed runs. Structured document and chart runs show much stronger visual sensitivity than the rationale-rich ScienceQA and A-OKVQA runs, but stronger drift does not automatically imply better answer accuracy.

ChartQA masking diagnostic: top-drift mask hurts the answer more than a random mask

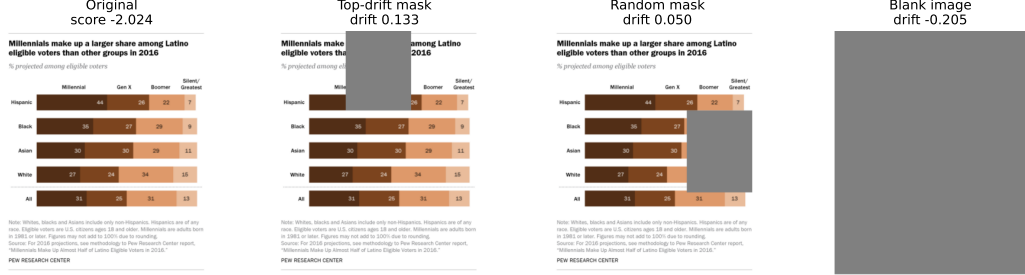


Figure 9: Example masking diagnostic for a ChartQA answer-only fallback run. The pipeline saves the original image, the top-drift mask, a random-region mask, and a blank-image control, together with the full-image and masked answer scores. This figure is used as an evidence-sensitivity example, not as proof that a generated rationale is faithful.

4.5. Efficiency and scale trade-offs

Table 11 and Figure 10 put Qwen2-VL-2B and Qwen2-VL-7B side by side on the same ScienceQA explanation-aware setting ($\alpha = 0.50$). Scaling up pays. The 7B run lifts accuracy from 0.765 to 0.885. ROUGE-L rises from 0.643 to 0.738, BLEU from 0.493 to 0.637. It is the single largest gain anywhere in the completed set. Both runs still fit under QLoRA. What the logs do not give us is a stable wall-clock or peak-memory figure, so this stands as a model-scale result, not a precise efficiency one.

Table 11: Qwen2-VL scale comparison on ScienceQA with the same explanation-aware objective ($\alpha = 0.50$).

Model	MC accuracy	ROUGE-L	BLEU
Qwen2-VL-2B	0.765	0.643	0.493
Qwen2-VL-7B	0.885	0.738	0.637

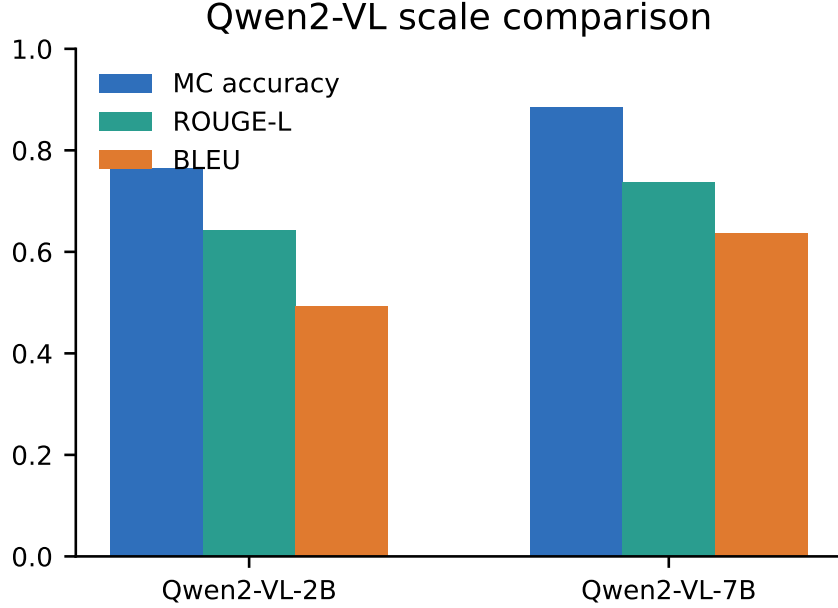


Figure 10: Scale comparison for the fixed ScienceQA explanation-aware setting. Moving from Qwen2-VL-2B to Qwen2-VL-7B improves both answer accuracy and rationale-overlap metrics.

4.6. Cross-domain transfer

Table 12 and Figure 11 lay out the train-domain against test-domain transfer matrix. Every column is scored in its target domain’s own metric, so the only safe comparisons run down a column, not across a row. Where the controls exist, the matrix keeps answer-only, rationale-generative, and explanation-aware sources separate. Treat it as an evaluation probe rather than the official in-domain table; a diagonal cell can drift a little from the matching completed run when the transfer evaluator routes through a different prompt family. The probe matters because answer-only training often travels better than rationale-generative training. The ScienceQA answer-only adapter is the clearest case, reaching 0.805 on A-OKVQA, 0.645 on ChartQA, 0.703 on DocVQA, and 0.630 on VQAv2. Its rationale-generative and explanation-aware siblings clear the random baselines but fall off on ChartQA, DocVQA, and VQAv2. A-OKVQA sources split the honours: rationale-generative training wins DocVQA transfer in that group at 0.764, while answer-only wins VQAv2 at 0.700. DocVQA is strongest at home (0.910), and the document-trained adapter carries over to ChartQA too

463 (0.685). The pattern, in short: transfer depends on the target domain and
 464 the source objective together, not on domain alone.

Table 12: Cross-domain transfer matrix. Rows indicate the training domain and columns indicate the evaluation domain. Each column uses the target domain’s headline metric, so values are best compared within a column.

Trained on	ScienceQA	A-OKVQA	ChartQA	DocVQA	VQAv2
ScienceQA answer-only	0.800	0.805	0.645	0.703	0.630
ScienceQA rationale CE	0.780	0.760	0.450	0.559	0.300
ScienceQA explanation-aware	0.740	0.765	0.460	0.491	0.410
A-OKVQA answer-only	0.690	0.825	0.635	0.677	0.700
A-OKVQA rationale CE	0.665	0.790	0.645	0.764	0.600
A-OKVQA explanation-aware	0.680	0.795	0.615	0.763	0.640
ChartQA answer-only	0.420	0.415	0.660	0.831	0.680
DocVQA answer-only	0.450	0.435	0.685	0.910	0.660
VQAv2 answer-only	0.440	0.575	0.660	0.716	0.730

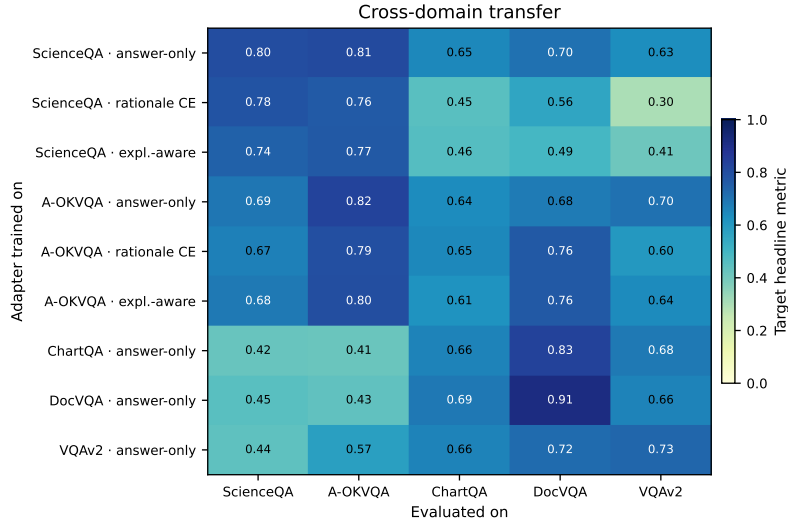


Figure 11: Cross-domain transfer heatmap. The strongest non-diagonal transfer is not always from the visually closest domain. Answer-only ScienceQA transfers strongly to A-OKVQA, while answer-only document and chart adapters transfer strongly to several open-ended targets.

465 5. Discussion

466 The results back the central idea, but only with caveats attached. They
467 also slot neatly into the arc the literature review traced: VLM alignment has
468 drifted from image–text representation matching toward generative instruc-
469 tion following, harder multimodal reasoning tests, and separate audits for
470 hallucination, critique, and evidence use. Objective choice clearly matters;
471 explanation supervision being automatically better than plain answer-only
472 fine-tuning does not follow. It can help, yes, but only when it is balanced with
473 care and held against a true answer-only control. On ScienceQA the best
474 fixed explanation-aware run is the low-answer-weight one, and the length-
475 aware control merely matches the rationale-generative control. The answer-
476 span-only setting is plainly worse on both accuracy and rationale quality.
477 A-OKVQA shows a small observed edge for explanation-aware training over
478 the generative control, but the capped-run uncertainty is wide enough that
479 we will not call it a firm win. And the fixed $\alpha = 0.50$ ScienceQA run is
480 actually worse than both the rationale-generative and answer-only controls.
481 Adding explanation tokens, by itself, does not make the model better. The
482 loss balance is what does the work.

483 The BLIP-2 contrastive result earns attention precisely because it is so
484 weakly positive on accuracy. We built the contrastive-enhanced arm specifi-
485 cally so the comparison could be run head-on. In the completed ScienceQA
486 run it sits just 0.005 above the BLIP-2 generative control, well inside the
487 approximate confidence intervals. Retrieval barely separates from the frozen
488 and random controls either. The read is that the auxiliary image–answer In-
489 foNCE term, at least in this small Q-Former-side setup, cannot forge a clearly
490 stronger image–answer representation. For this experiment set, generative
491 and explanation-aware supervision rest on firmer ground than contrastive
492 enhancement.

493 The domain results are the clearest argument against trusting a single
494 benchmark. ChartQA, DocVQA, VQAv2, A-OKVQA, and ScienceQA every
495 one improves in the selected runs. Yet the reading stays mixed, because the
496 domains are not all testing the same kind of supervision. ScienceQA and A-
497 OKVQA carry rationale-bearing targets; ChartQA, DocVQA, and VQAv2
498 fall back to answer-only training, because no gold rationales are there to
499 use. The strong DocVQA number, then, is capped extractive document
500 answering, nothing more, not evidence of faithful document rationales and
501 not a public-test DocVQA claim. Any stronger document-reasoning claim

would first need document-specific rationale supervision, or a more causal OCR and layout faithfulness test.

The faithfulness numbers come with a warning attached. High masking drift proves the model is sensitive to visual change; it does not prove the explanation is faithful. DocVQA and ChartQA drift hard because they live on local visual evidence, and in this run both also improve under plain answer-only fallback. ScienceQA and A-OKVQA post better answer and rationale scores, yet their drift stays small and refuses to climb consistently with explanation-aware training. Better rationale text, in other words, is not the same animal as more faithful visual grounding. So the masking diagnostic earns a place beside BLEU and ROUGE-L, never a place instead of them.

Scale hands us the cleanest positive result of all. Holding the ScienceQA explanation-aware objective fixed, Qwen2-VL-7B beats the 2B model on both accuracy and rationale metrics. That does not retire the 2B experiments, since the small model is exactly what made the wide sweep affordable in the first place. What it does say is that some of the 2B weakness is probably about capacity, not objective.

Transfer is mixed in the same way. Answer-only ScienceQA jumps cleanly to A-OKVQA, while the rationale-generative and explanation-aware ScienceQA adapters lose ground on the open-ended targets. A-OKVQA repeats the shape: rationale-bearing objectives help on some targets, but answer-only is the stronger bet for VQAv2. The answer-only ChartQA and DocVQA adapters trade well with each other under the target metrics, a hint that structured, text-heavy images share something useful when it comes to adaptation. The conclusion is the same one again: transfer rides on the target domain and the source objective together.

5.1. Limitations

Compute is the first limitation, and the obvious one. A single-GPU budget keeps the study realistic, but it also caps the seeds, the model sizes, and the full-dataset runs we could attempt. Rationale availability is the second. ScienceQA and A-OKVQA ship gold explanations; DocVQA, ChartQA, and VQAv2 do not, so they cannot carry the same rationale supervision, and any teacher-generated rationale experiment on them must be reported apart from gold-rationale training. The metrics are the third. BLEU and ROUGE-L can wave off a perfectly valid explanation just because the wording differs, and while the masking metric gets closer to actual evidence use, it hinges on

539 which regions are masked and turns noisy on dense documents and charts.
 540 The protocol is the fourth: capped train and evaluation subsets, one seed
 541 per full configuration. That is what makes the work feasible and auditable,
 542 but it also means small gaps are not statistically settled. And finally, the
 543 execution logs are good enough to validate the outputs, yet not good enough
 544 for precise wall-clock or peak-memory benchmarking. Taken together, these
 545 are the lines the paper will not cross in what it claims.

546 5.2. Future directions

547 The first thing future work should do is rerun the strongest configura-
 548 tions with more seeds and larger evaluation splits. After that, swap in newer
 549 open multimodal backbones while keeping the objective-isolation protocol
 550 untouched. A third push is better rationale annotation for chart and doc-
 551 ument datasets, the very places where faithful explanations would pay off
 552 most. The faithfulness test itself should grow more causal, editing document
 553 text, chart marks, or image regions in controlled ways rather than mask-
 554 ing blindly. There is also a clean comparison waiting between supervised
 555 explanation-aware training and preference-based or reinforcement-learning
 556 methods that reward grounding directly. And before any of this ships, de-
 557 ployment work should track latency, memory, and calibration alongside ac-
 558 curacy and faithfulness.

559 6. Conclusion

560 This paper asked what the alignment objective alone does to vision-
 561 language reasoning under a fixed, resource-constrained fine-tuning bud-
 562 get. The comparison spanned zero-shot baselines, BLIP-2 generative and
 563 contrastive-enhanced alignment, and Qwen2-VL in answer-only, rationale-
 564 generative, and explanation-aware forms, across document, chart, science,
 565 commonsense, and natural-image benchmarks. Throughout, we mea-
 566 sured answer accuracy, rationale quality, and evidence sensitivity together
 567 rather than one at a time. What holds up best in the completed runs is
 568 Qwen2-VL task adaptation: rationale supervision applied with care where
 569 gold rationales exist, answer-only fallback where they do not. BLIP-2
 570 contrastive-enhanced training is here as a route we genuinely tested, but
 571 its gain over BLIP-2 generative training is small and its retrieval diagnostic
 572 is weak. All five datasets, ScienceQA, A-OKVQA, ChartQA, DocVQA,
 573 and VQAv2, improve under the selected runs. The faithfulness analysis

lands on a distinction worth keeping: better explanation text and stronger visual evidence use are related, not identical. Explanation-aware training lifts rationale text in some settings, yet it does not reliably raise masking-based evidence dependence. Scale is the cleanest win, with Qwen2-VL-7B strengthening the ScienceQA explanation-aware result, while the transfer matrix shows alignment staying stubbornly target-domain dependent. The honest summary is a careful one: supervised rationales can help resource-constrained VLMs, but only when answer-only controls, domain shift, and the evidence test are weighed together.

Declarations

Author responsibility. The authors wrote, reviewed, and edited the manuscript and take full responsibility for the final content, results, and claims.

Use of generative AI. Generative AI tools were used only for language editing, code assistance, and consistency checks. All scientific decisions, experiment selection, result interpretation, and final manuscript approval remain the responsibility of the authors.

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Data, code, and artifacts. The experiments use public datasets cited in the paper. The code, configuration files, generated tables, figures, execution logs, and claim-to-evidence artifacts are maintained with the project repository and can be made available with the submitted supplementary material.

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